Design of Beam Diagnostic Equipment for Electron Sources Laboratory

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Abstract

The ongoing plan to upgrade the current Canadian Light source to a fourth generation synchrotron has multiple parts and one of them is to use new high energy electron sources in place of the current 220 keV Haimson gun. The aim of this project is to plan the setup for the new electron sources laboratory in order to charecterize the electron guns that would later be used to upgrade the synchrotron. This laboratory mainly comprises of an electron beam optics experimental setup which would help calibrate the energy, emittance and other features of the electron beam coming from the gun.

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1 Introduction

The new electron sources laboratory will be housed in an area near the LINAC. It will be mainly used to test the characteristics of electron guns which will be used to replace the current electron gun and also for the ultra low emittance fourth generation light source. The experiment comprises of two main parts: a quadrupole scanner and a dipole bender both of which are assembled using the spare quadrupoles and dipole which were previously used in the storage ring.

The diagnostic equipment is planned as follows: a solenoid or a focusing pair of quadrupoles after the electron gun followed by an aperture. This is later followed by a quadrupole triplet scanner and a dipole bender. A beam dump is placed at the tail end which can be used to measure the beam current.

The characteristics of each diagnostic tool is explained in detail in the following sections and an example simulation for the MAX IV RF gun is being used to test the beam diagnostics.

2 Solenoid

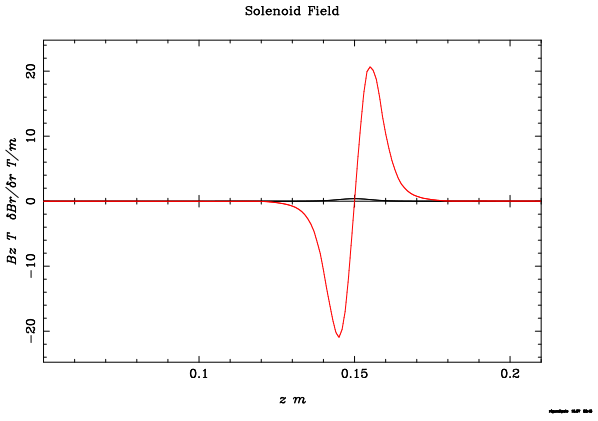


Figure 1: Solenoid Fields

In order to focus the beam right after the electron gun, a solenoid will be placed followed by an aperture. This solenoid needs to be exactly aligned with the longitudinal axis of the beam. This can be carried out either by a motorized system that can be controlled externally or a manual arrangement system that requires regular in-person monitoring during the setup of the layout. The solenoid fields also brings into force an xy correlation in the beam due to the induction of angular momentum to the beam. It is followed by an aperture and steering magnets followed by a beam position monitor in order to ensure the exact alignment of the electron beam.

3 Quadrupole Scanner

A quadrupole scanner can be used to measure the transverse emittance of the electron beam to a high accuracy. This is done by conforming to the fact that the beam optics can be described in terms of transfer matrices and if the distribution of the beam after the optics is known in position space, the entire phase space can be reconstructed by taking multiple measurements and finding the solutions to the transfer matrix equations.

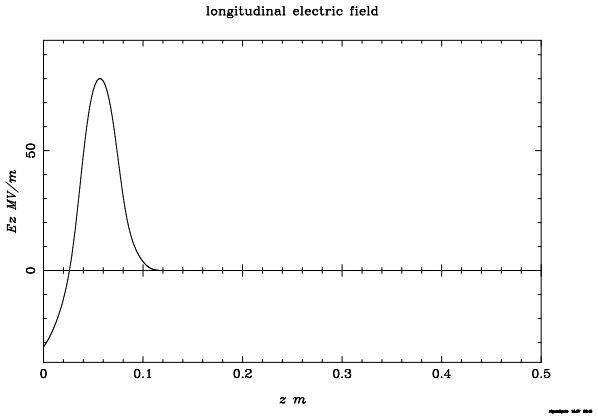


Figure 2: RF cavity fields in the MAX IV electron gun

In this case, we use three quadrupoles along the path of the beam and a beam profile monitor after an optimal drift length. By varying the quadrupole strength and collecting multiple measurements of the transverse beam size, we can reconstruct the transverse phase space and the emittance of the beam. This quadrupole scanning technique is called multiple-optics single-location scheme [2].

3.1 Theory

The transverse statistical properties which can be described entirely by the phase space elements can be written in the form of a 4D transverse matrix given by:

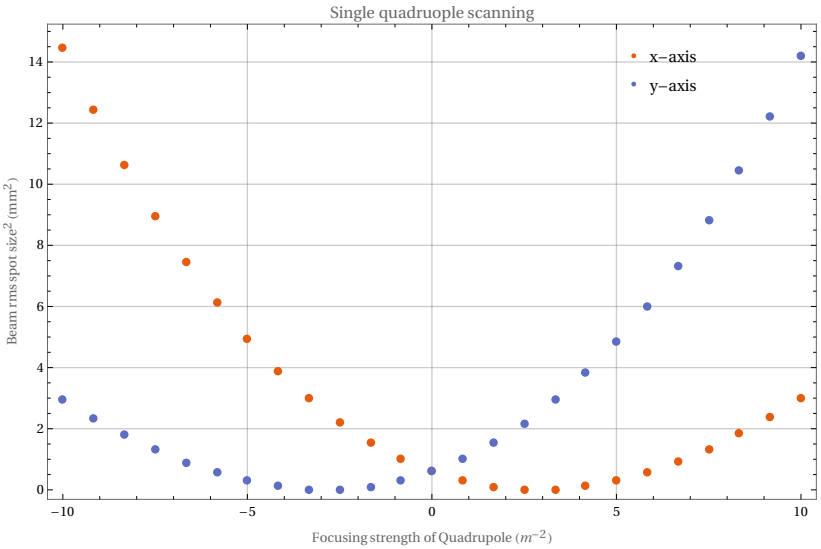


Figure 3: Variation of beam rms spot size in along the x and y direction with changing focusing strength of a single quadrupole scanner.

$$\sigma^{4D} = \begin{pmatrix} \langle x^2 \rangle & \langle xx' \rangle & \langle xy \rangle & \langle xy' \rangle \\ \langle xx' \rangle & \langle x'^2 \rangle & \langle x'y \rangle & \langle x'y' \rangle \\ \langle xy \rangle & \langle x'y \rangle & \langle y^2 \rangle & \langle yy' \rangle \\ \langle xy' \rangle & \langle x'y' \rangle & \langle yy' \rangle & \langle y'^2 \rangle \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy}^T & \sigma_{yy} \end{pmatrix} \tag{1}$$

Where x and y are horizontal and vertical coordinates and x' and y' are their derivatives with respect to the longitudinal coordinate. In the ideal case when x and y coordinates are decoupled, $\sigma_{xy} = 0$, and only measuring the 2D beam matrices would suffice to characterize the transverse emittance of the beam. But in reality, when the space charge forces provide non-negligible effect, there will be some xy correlations and $\sigma_{xy} \neq 0$. If we tend to ignore the σ_{xy} terms and only measure the 2D transverse emittances, then the obtained emittances are called *apparent* emittances.

Since the beam diagnostic apparatus is expected to be used for both low energy and high energy beams, the apparent emittances cannot be relied upon and necessitates the need to calculate the entire 4D matrix.

As the beam progresses along the quadrupole triplet, the transverse matrix changes according to the following relation:

$$\sigma_f = R\sigma_i R^T \tag{2}$$

Where R is the transport matrix between initial point i and final point f whose elements are determined by the quadrupole strengths, length and the drift space.

Once $\langle x^2 \rangle$, $\langle xy \rangle$ and $\langle y^2 \rangle$ are known, σ_{s_0} can be reconstructed by taking a series of measurements for different R matrices which gives us the transverse emittance of the beam right after the aperture.

The transfer matrix of the quadrupole triplet setup [4] can be calculated as follows:

For horizontally focusing quadrupole ($k < 0$), where k is the quadrupole strength and $\Omega = \sqrt{|k|}s$:

$$R_{QF} = \begin{pmatrix} \cos\Omega & \frac{1}{\sqrt{|k|}}\sin\Omega & 0 & 0 \\ -\sqrt{|k|}\sin\Omega & \cos\Omega & 0 & 0 \\ 0 & 0 & \cosh\Omega & \frac{1}{\sqrt{|k|}}\sinh\Omega \\ 0 & 0 & \sqrt{|k|}\sinh\Omega & \cosh\Omega \end{pmatrix} \tag{3}$$

Vertically focusing quadrupole ($k > 0$):

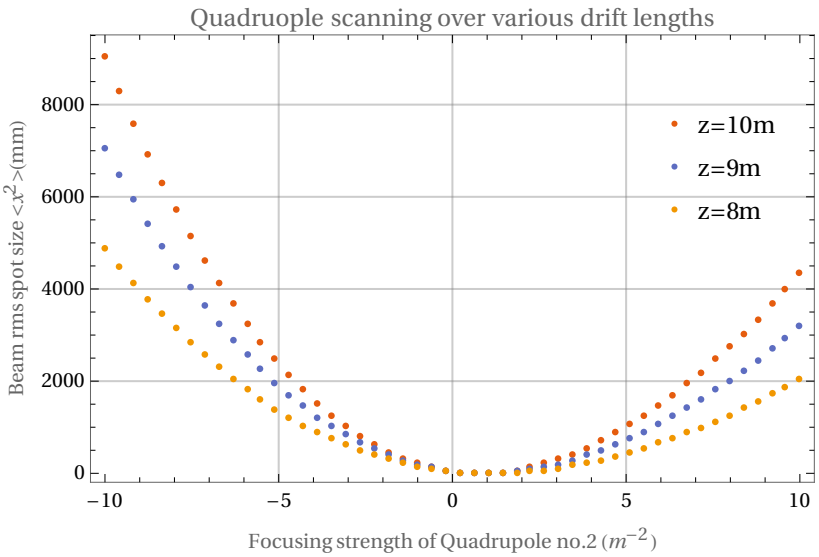


Figure 4: Variation of beam rms spot size in along the x direction with changing focusing strength of the 2nd quadrupole taken for different drift lengths.

$$R_{QD} = \begin{pmatrix} \cosh\Omega & \frac{1}{\sqrt{k}}\sinh\Omega & 0 & 0 \\ \sqrt{k}\sinh\Omega & \cosh\Omega & 0 & 0 \\ 0 & 0 & \cos\Omega & \frac{1}{\sqrt{k}}\sin\Omega \\ 0 & 0 & -\sqrt{k}\sin\Omega & \cos\Omega \end{pmatrix} \tag{4}$$

Drift region ($k = 0$) with s being the drift length:

$$D = \begin{pmatrix} 1 & s & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & s \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{5}$$

In our case, the lattice order from the aperture to the beam position monitor is as follows:

$$D_1\, R_{QF1}\, D_2\, R_{QD}\, D_3\, R_{QF2}\, D_4 = R \tag{6}$$

After multiplying the lattice elements, R be expressed in the following form:

$$R = \begin{pmatrix} R_{11} & R_{12} & 0 & 0 \\ R_{21} & R_{22} & 0 & 0 \\ 0 & 0 & R_{33} & R_{34} \\ 0 & 0 & R_{43} & R_{44} \end{pmatrix} \tag{7}$$

Now, using Eq.(2), we solve for the σ_i matrix elements by taking multiple measurements and solve the following equations:

$$\langle x^2\rangle_f = R_{11}^2\langle x^2\rangle_i + R_{12}^2\langle x'^2\rangle_i + 2R_{11}R_{12}\langle xx'\rangle_i \tag{8}$$
$$\langle y^2\rangle_f = R_{33}^2\langle y^2\rangle_i + R_{34}^2\langle y'^2\rangle_i + 2R_{33}R_{34}\langle yy'\rangle_i \tag{9}$$
$$\langle xy\rangle_f = R_{11}R_{33}\langle xy\rangle_i + R_{33}R_{12}\langle x'y\rangle_i + R_{11}R_{34}\langle y'x\rangle_i + R_{12}R_{34}\langle y'x'\rangle_i \tag{10}$$

There are 10 unknown elements here and 3 equations. Taking atleast 4 measurements by changing the quadrupole strengths, hence changing the R matrix elements, we can solve for the 10 elements and reconstruct the transverse emittance and Twiss parameters of the beam.

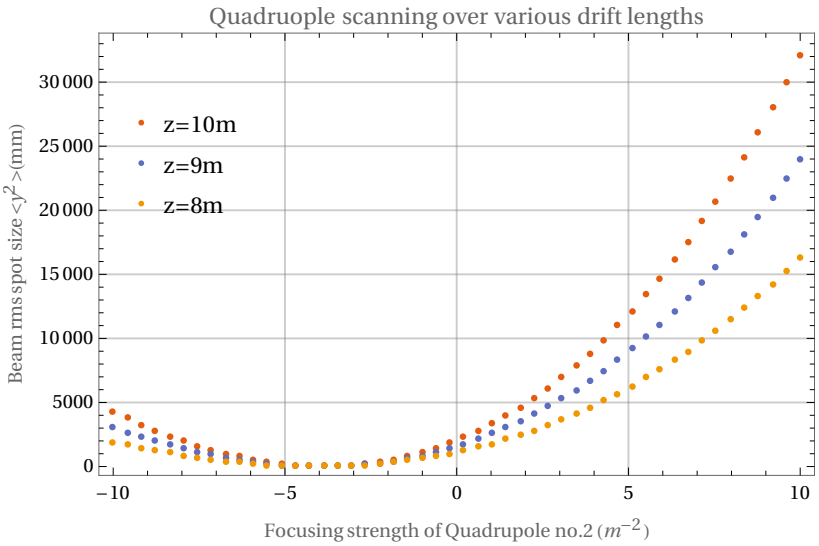


Figure 5: Variation of beam rms spot size in along the y direction with changing focusing strength of the 2nd quadrupole taken for different drift lengths.

In the case of the MAX IV RF gun, the $\langle xy \rangle$ terms are negligible, hence the last part of the previous set of equations can be ignored. So, this brings it down to just finding the 2D Twiss parameters for the individual coordinates. Which leaves us with just the these two equations:

$$\langle x^2 \rangle_f = R_{11}^2 \langle x^2 \rangle_i + R_{12}^2 \langle x'^2 \rangle_i + 2R_{11}R_{12} \langle xx' \rangle_i \tag{11}$$

$$\langle y^2 \rangle_f = R_{33}^2 \langle y^2 \rangle_i + R_{34}^2 \langle y'^2 \rangle_i + 2R_{33}R_{34} \langle yy' \rangle_i \tag{12}$$

Now that x and y are independent, the two equations can be solved for seperately by taking atleast 3 different measurements.

For simplicity, let:

$$R_{11}^2 = a, \quad R_{12}^2 = b, \quad 2R_{11}R_{12} = c$$

Now for three different quadrupole strengths we will have the following set of equations:

$$\begin{pmatrix} \langle x^2 \rangle_1 \\ \langle x^2 \rangle_2 \\ \langle x^2 \rangle_3 \end{pmatrix} = \begin{pmatrix} a1 & b1 & c1 \\ a2 & b2 & c2 \\ a3 & b3 & c3 \end{pmatrix} \begin{pmatrix} \langle x^2 \rangle \\ \langle x'^2 \rangle \\ \langle xx' \rangle \end{pmatrix} \tag{13}$$

Hence,

$$\begin{pmatrix} \langle x^2 \rangle \\ \langle x'^2 \rangle \\ \langle xx' \rangle \end{pmatrix} = \begin{pmatrix} a1 & b1 & c1 \\ a2 & b2 & c2 \\ a3 & b3 & c3 \end{pmatrix}^{-1} \begin{pmatrix} \langle x^2 \rangle_1 \\ \langle x^2 \rangle_2 \\ \langle x^2 \rangle_3 \end{pmatrix} \tag{14}$$

Which will provide us with the required twiss parameters of the beam at the point of reconstruction. The normalized rms emittance is expressed as:

$$\epsilon_{x,rms} = \beta\gamma \sqrt{\langle x^2 \rangle \langle x'^2 \rangle - \langle xx' \rangle} \tag{15}$$

The twiss parameters are given by:

$$\beta_x = \langle x^2 \rangle / \epsilon_x, \quad \gamma_x = \langle x'^2 \rangle / \epsilon_x, \quad \alpha_x = -\langle xx' \rangle / \epsilon_x \tag{16}$$

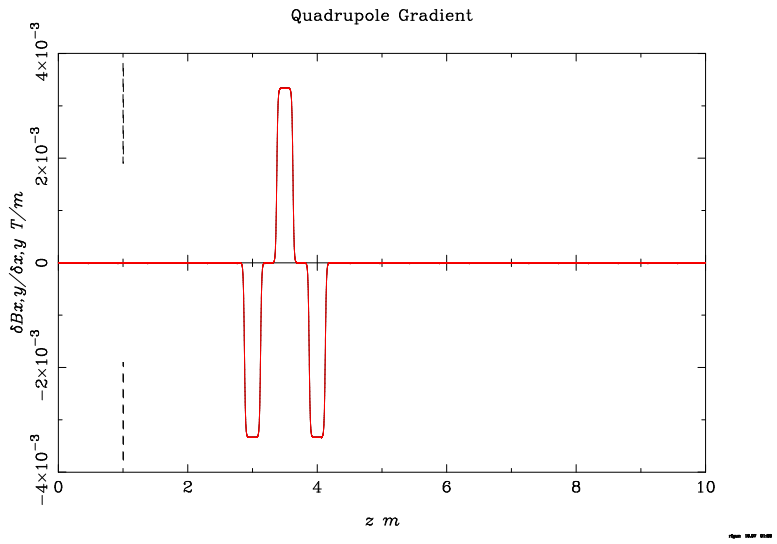


Figure 6: Quadrupole triplet field gradient. Note the presence of aperture at $z = 1$m.

The above exercise yields exact results only when the space charge effects are negligible. But in the energy range of a few keV to MeV, the space charge forces do play a prominent role in increasing the transverse emittance as the beam progresses along the z direction. To take this into consideration while reconstructing the transverse emittance of the beam, we would have to find the fit parameters to the parabola separately and eventually find the twiss parameters.

Making use of a quadrupole triplet here is advantageous over using a single quadrupole since by varying the strengths of different quadrupoles, multiple parabolas can be fit and more precise transverse emittance measurements can be obtained. Also, it can be more compatible when a wide range of energy electron guns are used and the quadrupole triplet strengths can be adjusted to cater to obtaining an optimal beam size at the point of beam size monitoring.

3.2 Simulations

In order to obtain an estimate of the quadrupole strengths and drift space lengths, simulations were being performed using DIMAD software [3].

Further simulations were being performed in ASTRA [1] in order to account for space charge effects and also fringe field effects in the magnets.

As a starting example, the ASTRA code of MAX-IV RF gun whose RF fields are shown in Fig(2) was being used. Here, the gun is followed by a focusing solenoid Fig.(1) which is followed by an aperture at a distance of 1m from the source. The code is extended with the addition of a quadrupole triplet lattice whose fields are shown in Fig. (6)

The lengths of the quadrupoles is fixed to be 0.25m. The quadrupoles allow for a wide range of field strength variation, hence their strengths will be varied to different degrees in order to optimize the method of scanning that can be used to measure the transverse beam emittance effectively. Firstly, fixing the drift distances and varying the field strength of just the second quadrupole, we get the following plots for the rms beam spot sizes. Then the same parameter space is scanned but by varying the final drift length as well (considering the constraint that the dipole bender would be placed after the quadrupoles triplet). Now looking into this 3D space of constraints, the optimal method of scanning to be used shall be decided.

3.2.1 Reconstruction of the transverse emittance

The point of reconstruction of the beam is chosen after a sufficient drift distance after the aperture in order to maintain the assumption of a symmetric Gaussian distribution of the electrons in the beam. The aperture is placed at $z = 1m$ The quadrupole triplet starts at $z = 2.5m$.

Firstly, let us just consider the simple case with just a single quadrupole scanner and varying the magnetic field gradient as shown in Fig.(3). The transverse emittance at the point of reconstruction ($z = 1.5m$) with the single quadrupole of length 0.25 m placed at $z = 3m$ is $\epsilon_x = 3.314\pi$ mrad mm and $\epsilon_y = 3.282\pi$ mrad mm (as obtained by Astra).

In order to test the reconstruction accuracy, the R matrix will be as follows(with the quadrupole parameters being $k = -3.27 \times 10^{-3}\, T/m$ and $l = 0.25m$):

$$R = D_1 R_{QF1} D_2 \tag{17}$$

$$D_1 = \begin{pmatrix} 1 & 1.375 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1.375 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{18}$$

$$R_{QF1} = \begin{pmatrix} \cos\Omega & \frac{1}{\sqrt{|k|}}\sin\Omega & 0 & 0 \\ -\sqrt{|k|}\sin\Omega & \cos\Omega & 0 & 0 \\ 0 & 0 & \cosh\Omega & \frac{1}{\sqrt{|k|}}\sinh\Omega \\ 0 & 0 & \sqrt{|k|}\sinh\Omega & \cosh\Omega \end{pmatrix} \tag{19}$$

$$D_2 = \begin{pmatrix} 1 & 6.875 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 6.875 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{20}$$

$$\tag{21}$$

Now reading off 3 values of rms beam size from Fig. (3), we obtain the reconstructed transverse emittances.

4 Dipole Bender

A dipole bender is used to measure the average energy and the energy spread of the beam. In this case a 45 deg dipole bender is placed after the quadrupole triplet setup. It is ideally operated when the quadrupole triplet is switched off.

4.1 Theory

The transfer matrix for the dipole bender taking into account the dispersion and momentum compaction factor are as follows:

$$M = \begin{pmatrix} \cos\frac{s}{R} & R\sin\frac{s}{R} & R(1-\cos\frac{s}{R}) \\ -\frac{1}{R}\sin\frac{s}{R} & \cos\frac{s}{R} & \sin\frac{s}{R} \\ 0 & 0 & 1 \end{pmatrix} \tag{22}$$

$$\begin{pmatrix} \langle x^2\rangle \\ \langle x'^2\rangle \\ \Delta p/p \end{pmatrix} = M \cdot \begin{pmatrix} \langle x_0^2\rangle \\ \langle x_0'^2\rangle \\ \Delta p/p \end{pmatrix} \tag{23}$$

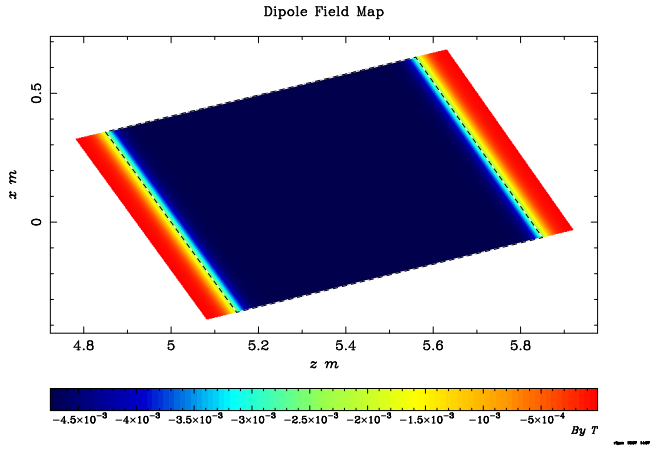


Figure 7: Dipole field simulated in ASTRA

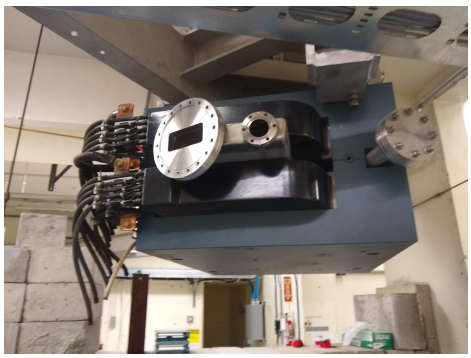


Figure 8: Dipole bender

5 Layout and Design

The dimensions of the setup is given in Table (1). The schematic is shown in Fig.(9)

6 Acknowledgements

I am thankful to Mark Boland for providing me this oppportunity to spend 12 weeks in Canada with the Mitacs scholarship. I also thank him for the enlightening discussions we had about accelerator physics. I thank Drew for providing me this interesting project statement and guiding me throughout with the software and helping me think better by asking a lot of questions. I am extremely thankful to Les Dallin for sparing his time to help me initiate the codes required for the beam optics and also guiding me with the plan to design the setup. I thank Allan for the several interesting discussions. I also thank the other members and summer students of CLS for making my stay a pleasant one.

References

[1] K. Floettmann. ASTRA, http://www.desy.de/ mpyflo/. March 2017.

[2] Eduard Prat and Masamitsu Aiba. Four-dimensional transverse beam matrix measurement using the multiple-quadrupole scan technique. *Phys. Rev. ST Accel. Beams*, 17:052801, May 2014.

[3] R. Servranckx, Karl L. Brown, Lindsay Schachinger, and David Douglas. Users Guide to the Program DIMAD. 1990.

Table 1: Layout dimensions

Arrangement of lattice along the longitudinal (z) direction	
Lattice element	Starting position z m
Solenoid	0.14
Drift 1	0.16
Aperture	0.99
Drift 2	1.0
Corrector magnets	1.05
Drift 3	1.15
Beam position monitor	1.16
Drift	1.26
Quadrupole 1	2.825
Drift 3	3.125
Quadrupole 2	3.375
Drift 4	3.625
Quadrupole 3	3.875
Drift 5	4.125
Dipole 1	4.85
Drift 6	5.85
Beam position monitor	9.9
Drift	10.0
Beam dump	10.1

[4] Klaus Wille. *The Physics of Particle Accelerators*. 2005.

7 Codes

```
 &NEWRUN
 Head='Max IV thermionic gun'
 RUN=1,
 Loop=F,
 Nloop=1,
 Distribution = 'rfgun.ini',
 Xoff=0.E-3,
 Yoff=0.0
 Qbunch=0.5,
 TRACK_ALL=.T,
 AUTO_PHASE=.F,
 PHASE_SCAN=.F,
 check_ref_part=.F,
 H_max=0.01,
 H_min=0.001,
 Lprompt=.F
/

&OUTPUT
 ZSTART=0,
 ZSTOP=10,
 Zemit=160,
 Zphase=25,
 Lmagnetized=.F,
 EmitS=.T,
 C_EmitS=.F,
 TR_EMITS=.F,
 PhaseS=.T,
 TrackS=.T,
 RefS=.T,
 TcheckS=.F,
 CathodeS=.F,
/

&SCAN
 LScan=T
 Scan_para='Q_K(2)',
 S_min=-10 , S_max=10 ,  S_numb=50
 !FOM(1)='momentum'
 FOM(1)='hor spot size'
 !FOM(2)='hor emit'
 !FOM(3)='hor div'
 FOM(2)='ver spot size'
 !FOM(2)='ver emit'
 !FOM(3)='ver div'
 !FOM(1)='bunch charge',
 !FOM(2)='hor emit'
 !FOM(3)='bunch length'
 !FOM(4)='hor spot size'
/

&CHARGE
 Loop=F
 LSPCH=T
 Nrad=10, Nlong_in=10
 Cell_var=2.0
```

```
 min_grid=0.4D-6
 Max_scale=0.05
 Max_count=100
 Lmirror=.F
/

&APERTURE
 LApert=T,
 File_Aperture(1)='rad',
 Ap_R(1)=1, Ap_Z1(1)=0, Ap_Z2(1)=0.005,
 A_Pos(1)=1
/



&CAVITY
 Loop=.T,
 LEFieLD=.T
 FILE_EFieLD(1) = 'gunfield.dat',
 Nue(1)= 2.856005, MaxE(1)=80.0,Phi(1)=30.0, C_pos(1)=0.0, C_smooth(1)=5
/


&SOLENOID
 LBField=.T
 FILE_BField(1) = 'solt2-3.4A.dat',
 MaxB(1)=0.38, S_pos(1)=0.15
/


&QUADRUPOLE
 LOOP=False
 Lquad=T
 !Q_K(1)=0.92092766606743
 Q_K(1)=0.98
 Q_length(1)=0.25
 !Q_dist(1) = 0.5
 Q_pos(1)=3
 Q_xoff(1)=0
 Q_yoff(1)=0
 Q_K(2)=-0.98
 Q_length(2)=0.25
 Q_pos(2)=3.5
 Q_xoff(2)=0
 Q_yoff(2)=0
 Q_K(3)=0.98
 Q_length(3)=0.25
 Q_pos(3)=4
 Q_xoff(3)=0
 Q_yoff(3)=0
/


&DIPOLE
 LOOP=FALSE
 LDipole=F
 D_Type(1)=hor
 D1(1)=(0.35,4.85)
```

```
 D2(1)=(-0.35,5.15)
 D3(1)=(0.64,5.56)
 D4(1)=(-0.06,5.85)
 D_Gap(1,1)=0.05
 D_Gap(2,1)=0.05
 D_radius(1)=1.0
/
```

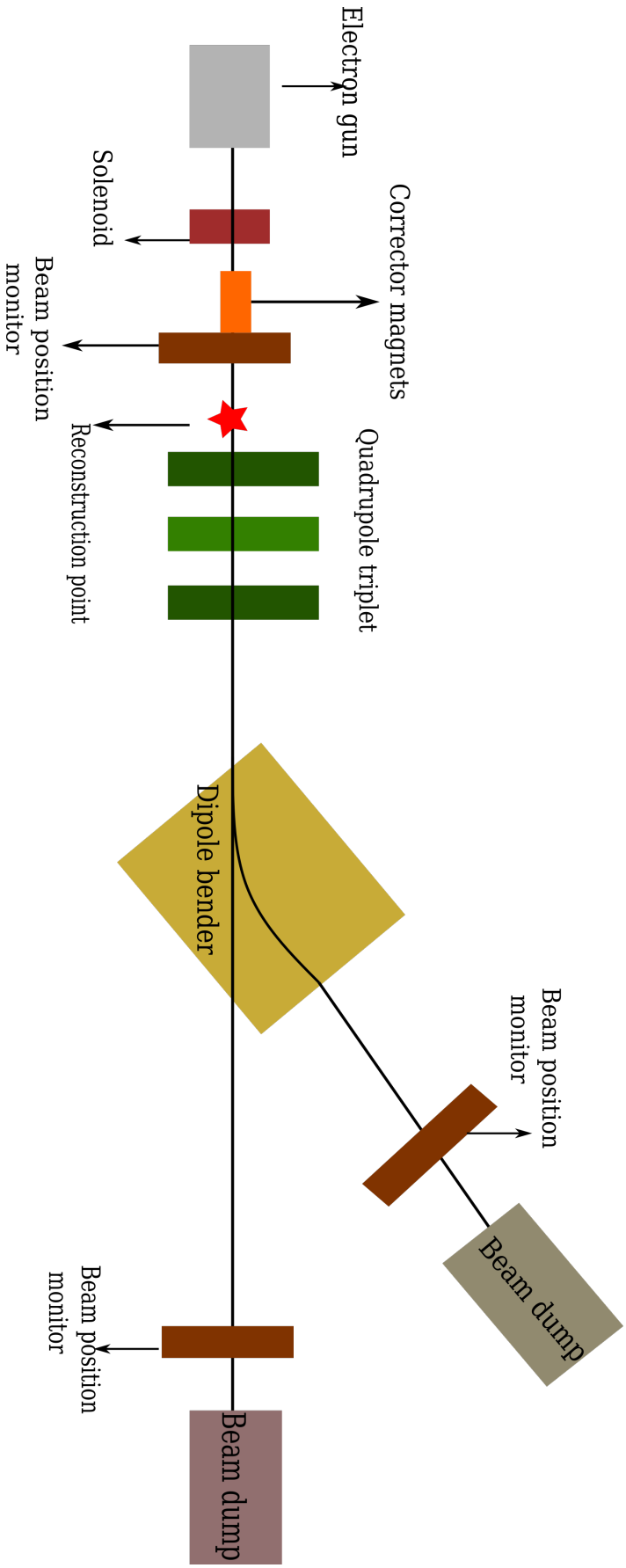


Figure 9: Layout of the entire beam diagnostic setup

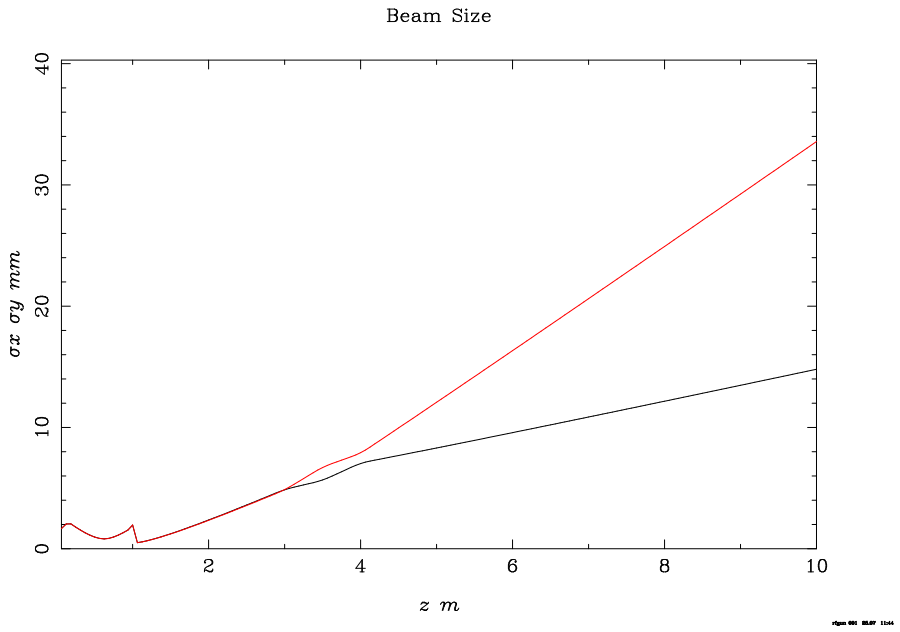


Figure 10: Variation of rms beam size as it passes through the quadrupole scanner

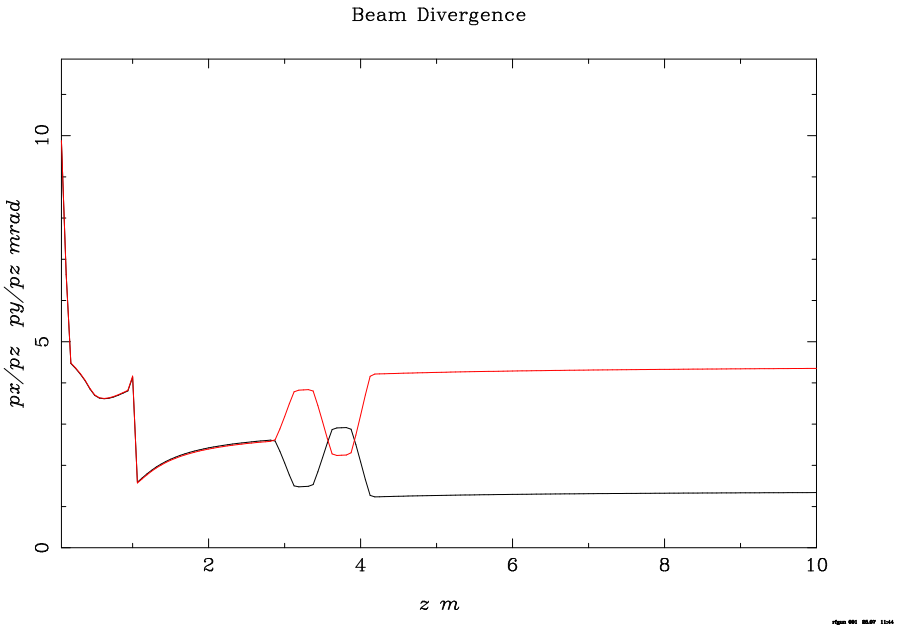


Figure 11: Variation of rms beam divergence as it passes through the quadrupole scanner

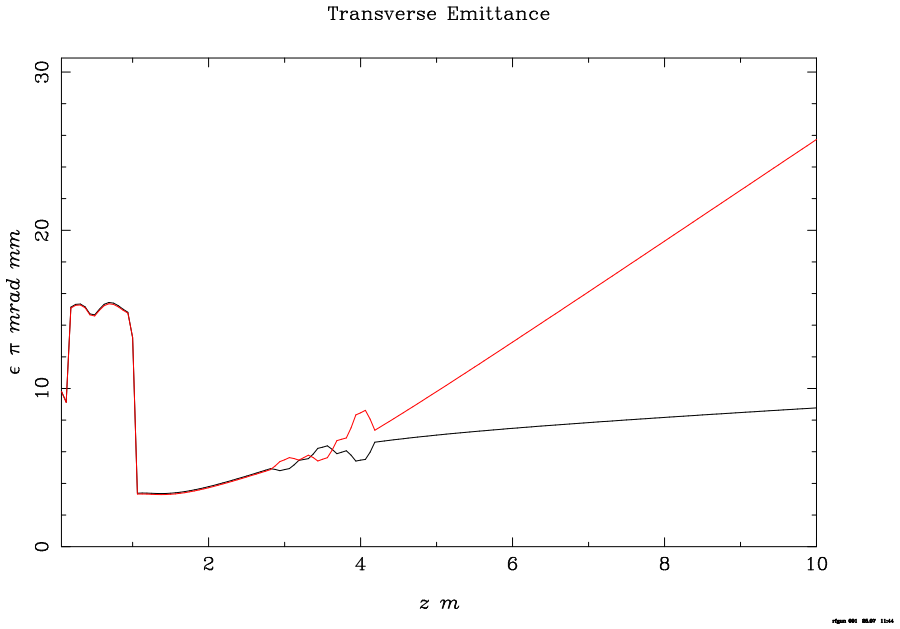


Figure 12: Variation of transverse emittance as it passes through the quadrupole scanner

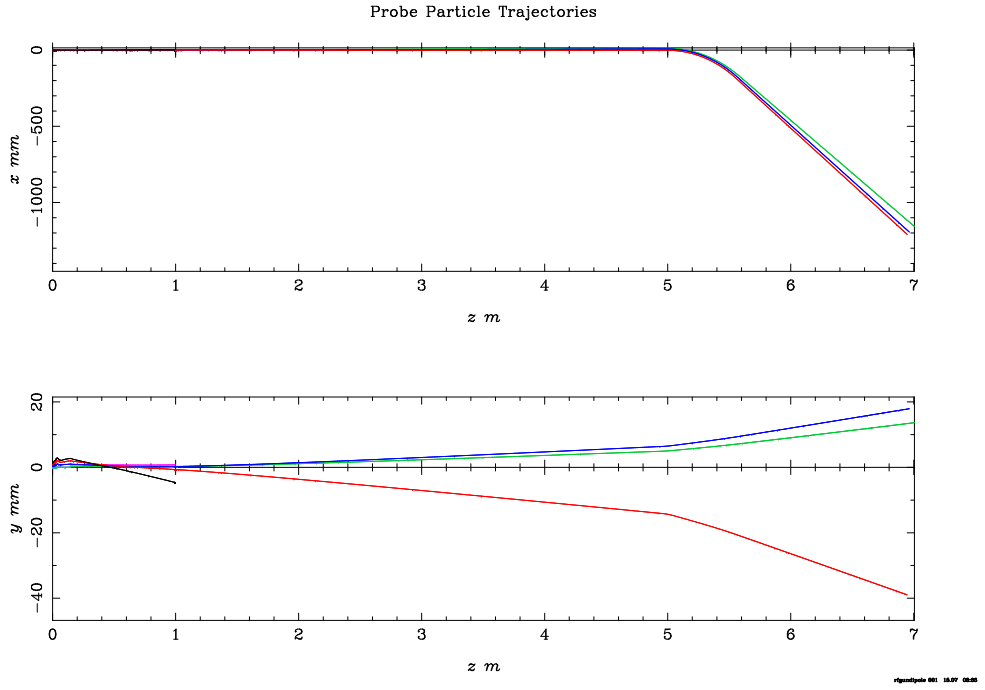


Figure 13